



PYTHON FOR AI PROGRAMMER MASTERCLASS

From Beginner to AI Developer – Build Intelligent Applications with Python

Master Python programming from the ground up and develop AI-powered applications using modern libraries, automation, machine learning, deep learning, APIs, and Generative AI. This course is designed for aspiring AI Engineers, Data Scientists, ML Developers, and GenAI Professionals.



Course Roadmap (Beginner → Advanced)

01. Python Fundamentals

- Python Installation & IDE Setup
 - Variables & Data Types
 - Operators & Expressions
 - Input & Output
 - Type Conversion
 - Comments & Documentation
 - Basic Programming Logic
 - Mini Practice Programs
-

02. Control Flow & Functions

- Conditional Statements
 - Loops (For & While)
 - Functions & Parameters
 - Return Values
 - Lambda Functions
 - Recursion
 - Error Handling
 - Code Reusability
-

03. Data Structures

- Lists
- Tuples
- Sets
- Dictionaries

- String Manipulation
 - List Comprehensions
 - Iterators & Generators
 - Collections Module
-

04. Object-Oriented Programming

- Classes & Objects
 - Constructors
 - Inheritance
 - Polymorphism
 - Encapsulation
 - Abstraction
 - Magic Methods
 - Dataclasses
-

05. File Handling & Automation

- Read & Write Files
 - CSV & JSON Processing
 - Excel Automation
 - OS Module
 - Pathlib
 - Logging
 - Scheduling Tasks
 - Email Automation
-

06. Python Libraries for AI

- NumPy
 - Pandas
 - Matplotlib
 - Seaborn
 - Plotly
 - SciPy
 - OpenCV Basics
 - Pillow
-

07. Machine Learning with Python

- Data Preprocessing
 - Feature Engineering
 - Scikit-learn
 - Regression
 - Classification
 - Clustering
 - Model Evaluation
 - Hyperparameter Tuning
-

08. Deep Learning & NLP

- TensorFlow
 - Keras
 - PyTorch Basics
 - Neural Networks
 - CNN Fundamentals
 - RNN & LSTM
 - Transformers Overview
 - Natural Language Processing
-

09. Generative AI Development

- Prompt Engineering
 - OpenAI API
 - Hugging Face
 - LangChain
 - LlamaIndex
 - RAG (Retrieval-Augmented Generation)
 - AI Agents
 - Multimodal AI Applications
-

10. Advanced AI Programming

- FastAPI for AI APIs
- Streamlit Dashboards
- Gradio Interfaces
- Vector Databases

- Docker for AI Apps
 - Git & GitHub
 - Cloud Deployment
 - AI Project Architecture
-



Hands-on AI Projects

- Python Automation Scripts
 - AI Chatbot
 - PDF Q&A Assistant
 - Resume Analyzer
 - AI Image Caption Generator
 - Sentiment Analysis System
 - Recommendation Engine
 - Document Summarizer
 - AI Code Assistant
 - End-to-End RAG Application
-



Tools & Technologies

- Python 3.x
 - VS Code
 - Jupyter Notebook
 - Google Colab
 - NumPy
 - Pandas
 - Matplotlib
 - Scikit-learn
 - TensorFlow
 - PyTorch
 - OpenAI API
 - Hugging Face
 - LangChain
 - FastAPI
 - Streamlit
 - Docker
 - Git & GitHub
-



Career Opportunities

- AI Programmer
 - Python Developer
 - AI Application Developer
 - Machine Learning Engineer
 - Data Scientist
 - Generative AI Engineer
 - Prompt Engineer
 - NLP Engineer
 - AI Automation Engineer
 - AI Solutions Architect
-



Course Highlights

- ☒ Beginner to Advanced Curriculum
 - ☒ Hands-on Coding Labs
 - ☒ Real-World AI Projects
 - ☒ Machine Learning & Deep Learning
 - ☒ Generative AI & LLM Development
 - ☒ AI Agents & RAG Applications
 - ☒ API Development & Deployment
 - ☒ Portfolio-Ready Projects
 - ☒ Interview Preparation
 - ☒ Industry Best Practices
-



Tagline

"Code with Python. Build with AI. Innovate for the Future."